Khushaan Virk

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*

- **Relevant Coursework:** Circuit Analysis, Digital Logic Design, Signals & Systems, Embedded Systems Design

Technical Skills

Embedded & Firmware: ARM (STM32F072), bare-metal C, USB HID, SWD bring-up, ADC, I2C, SPI, UART, PWM
PCB Design: Altium Designer, KiCad, 4-layer stackup, controlled-impedance routing, DRC/DFM, JLCPCB fab and assembly
Programming: Python, C, Embedded C, SystemVerilog, SQL
Tools & Instrumentation: Git, OpenOCD/GDB, ST-Link, oscilloscopes, logic analyzers, SMD soldering, MATLAB

Hardware & Firmware Projects

65% Mechanical Keyboard PCB | *Altium, STM32F072, 4-Layer, Bare-metal C, USB HID* | writeup — Aug. 2026

- Retuned Altium's solder-mask rule to JLCPCB's real 0.1mm capability, which cut 194 false DRC violations to 2 and surfaced a diode 0.152mm off grid that would have shipped as a dead key
- Derived the USB differential pair by hand from IPC-2141 rather than a vendor calculator and routed it at 0.30/0.20mm for 91.0 Ω against a 90 Ω target, top layer over solid ground. The board worked on the first spin, 67 of 67 keys
- Brought the board up over SWD with OpenOCD and GDB before pressing a key. SysTick at 2023 ticks in two seconds proved the 48 MHz clock, and a tweezer sweep recovered the matrix map from silicon
- Rewrote the firmware bare-metal once QMK had validated the hardware: no HAL, no CubeMX, CMSIS startup and a hand-written Makefile, 1 kHz matrix scan, 5 ms deferred debounce, USB HID via TinyUSB

Pedestrian Dead Reckoning (Phone IMU) | *Python, NumPy, SciPy, pandas* | writeup — Jul. 2026

- Built a Python signal-processing pipeline in NumPy and SciPy to reconstruct a walking path from phone accelerometer and gyroscope logs alone, band-passing acceleration magnitude to the 0.7–2.1 Hz gait band and running peak detection for footfall counting, with cadence cross-checked against an FFT of the walk
- Estimated gyroscope bias from a stationary window, integrated the corrected rate for heading, and dead-reckoned an (x, y) trajectory step by step, measuring inertial drift by loop closure over a 76.2 m route at 5.1 to 6.0 m, roughly 7% of distance walked

Field-Following Autonomous Robot | *STM32, C, MOSFETs, I2C, IR* | writeup — Apr. 2026

- Built a dual-microcontroller robot with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Integrated VL53L0X Time-of-Flight laser ranging modules over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *Op-Amp, ADC, SSR, FSM, Python* | writeup — Feb. 2026

- Designed a closed-loop temperature control system using a K-type thermocouple with cold-junction compensation (LM335) and OP07 op-amp signal conditioning, achieving ±3°C accuracy over 25–240°C
- Built a Python serial data acquisition framework around the firmware: real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud, used as the verification harness during firmware tuning

Experience

Software Engineer — Jan. 2026 – Present
Advizr — *Vancouver, BC*

- Own uQueue, a Canada-wide clinic appointment search built on a .NET API, an Angular front end and a Python scraper, and moved it onto Advizr's Railway and Supabase infrastructure across 3,036 Jane subdomains in BC
- Diagnosed a go-live outage where the scraper sent every database call to asyncio's unbounded thread pool and drained the 15-connection Supabase session pooler it shared with the .NET backend. Capped it at 5 threads
- Traced 24 to 28 hour scrape cycles to roughly 47,000 routine 404s a run, each circuit-breaker trip parking a host for 300 seconds. Reclassified them as authoritative empty results and added a per-URL deadline

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025
City of Surrey — *Surrey, BC*

- Led structured recreation programs for groups of 15–30, coordinating daily operations across staff and volunteers